Conditional Survival

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Introduction

Conditional survival is the probability of surviving an additional time t given that a patient has already survived to time s. It answers the clinically relevant question that absolute survival cannot: a five-year survival probability of 30 % at diagnosis says little to a patient who has already lived three years. The conditional five-year-from-now estimate may be 70 % or higher, dramatically updating the prognosis. Survivorship clinics, oncology long-term follow-up, and patient counselling all rely on conditional rather than unconditional survival.

Prerequisites

A working understanding of the survival function $S(t)$, conditional probability, and the Kaplan-Meier estimator.

Theory

By the definition of conditional probability,

$$\mathrm{P}(T > t + s \mid T > s) = \frac{\mathrm{P}(T > t + s)}{\mathrm{P}(T > s)} = \frac{S(t+s)}{S(s)}.$$

A patient who has survived s years is now in the risk set at s; their next-t-year survival is the ratio of survival probabilities. For Kaplan-Meier estimates, this is computed from $\hat{S}(s)$ and $\hat{S}(t+s)$ directly. Confidence intervals require variance propagation through the ratio; the log or log-log transformation gives intervals that respect the $[0, 1]$ bounds.

Assumptions

A reliable Kaplan-Meier or parametric survival estimate at both s and $t + s$; in practice this means sufficient follow-up so that $\hat{S}(t+s)$ is not based on a tiny remaining cohort.

R Implementation

```r
library(survival)

data(lung)
fit <- survfit(Surv(time, status) ~ 1, data = lung)

# Unconditional 2-year survival
summary(fit, times = c(500, 1000))$surv   # approximate 2- and 3-year

# Conditional 1-year survival given 1-year survivorship
times <- summary(fit, times = c(365, 730))
cond_surv <- times$surv[2] / times$surv[1]
cond_surv
```

Output & Results

The ratio of two KM survival estimates at chosen landmark times. For confidence intervals, transform onto the log-log scale, compute SEs by the delta method, and back-transform. The `condsurv` package automates the computation including bootstrap intervals.

Interpretation

A reporting sentence: "Unconditional one-year survival was 48 %; conditional on having survived one year, the probability of surviving an additional year was 66 % (95 % CI 56 % to 75 %); conditional on having survived two years, the next-year survival rose to 78 %." Always report the landmark time s explicitly; conditional survival is meaningless without it.

Practical Tips

- Report conditional survival at clinically meaningful landmarks: one-year, five-year, ten-year survivor checkpoints in oncology, post-transplant 90-day survival in transplantation.
- `condsurv` automates the computation and produces clean tables and plots; it handles confidence intervals via the log-log transformation by default.
- Confidence intervals must respect the $[0, 1]$ bounds; the log-log transformation is the standard choice and is preferable to the naive Wald interval.
- For dynamic prediction (conditional survival as a function of s), landmark analyses or joint longitudinal-survival models give more refined estimates that incorporate post-baseline information.
- Conditional survival is particularly useful in patient counselling because it answers the question patients actually ask: "Given that I'm still alive, how much longer can I expect to live?" rather than the population-level survival reported at diagnosis.
- Pair conditional-survival reports with the underlying KM curve so readers can see how the conditional estimate evolves across the cohort's experience.

R Packages Used

`survival` for the Kaplan-Meier baseline; `condsurv` for tabular conditional survival with CIs; `survminer::ggsurvplot()` for KM curves to display alongside conditional summaries; `dynpred` and `JM` for dynamic prediction beyond the simple ratio of survival probabilities.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation